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Method and system for predicting content based recommendations

Abstract

The embodiments herein provide a method for predicting content based recommendations. The method includes recognizing, by a first electronic device (**100A**) in a first IoT network (**100**), content displayed on a display (**130A**) of the first electronic device (**100A**). Further, the method includes predicting, by the first electronic device (**100A**), at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device (**100A**) and other electronic devices (**100B-100F**) in the first IoT network (**100**). Further, the method includes providing, by the first electronic device (**100A**), the at least one recommendation on an action prediction panel of a keyboard of the first electronic device (**100A**).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a 371 of International Application No. PCT/KR2019/015778 filed on Nov. 18, 2019, which claims priority to India Patent Application No. 201841043507 filed on Nov. 19, 2018 and India Patent Application No. 201841043507 filed on Nov. 5, 2019, the disclosures of which are herein incorporated by reference in their entirety.

BACKGROUND

1. Field

(2) The present disclosure relates to electronic devices, and more specifically to a method and system for predicting content based recommendations.

2. Description of Related Art

(3) Electrical appliances connect to a network to form an Internet of Thing (IoT) device environment, where a user can monitor and control operations of the electrical appliances using an electronic device such as smartphone, laptop etc. Conventional systems allow the user to control the operations of the electrical appliances in the IoT device environment based on a voice command received from the user. Further, the conventional systems predict an activation of an operation of the electrical appliances based on a history of user activities performed using the electrical appliances. Example for the user activities performed using the electrical appliances are watching a video in a smart television (TV), doing workout on a treadmill, etc.

(4) In an example scenario, a user is watching the video in a first smart TV, where a control configuration of the first smart TV to watch the video includes an application used to play the video, a link of the video to fetch the video from a server, a sound level of the first smart TV in playing the video, etc. However, the conventional systems lack in sharing the control configuration of the smart TV to other electrical appliances such as a second smart TV, a smart phone etc., which are capable of applying the control configuration to play the video.

(5) The conventional systems allow the user to manually share the link of the video, details about the application used to play the video, etc. to the other electrical appliances. However, recommending the control configuration to the user for sharing the control configuration to other electrical appliances is desirable. Thus, it is desired to at least provide a useful alternative.

(6) The principal object of the embodiments herein is to provide a method and system for predicting content based recommendations.

(7) Another object of the embodiments herein is to recognize content displayed on a display of a first electronic device in a first IoT network.

(8) Another object of the embodiments herein is to generate at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic device in the first IoT network.

(9) Another object of the embodiments herein is to provide the at least one recommendation to the user for selecting at least one recommendation.

(10) Another object of the embodiments herein is to automatically include the at least one selected recommendation to compose a message for a user of a first electronic device in a second IoT network.

(11) Another object of the embodiments herein is to determine at least one electronic device in the second IoT network, relevant to the at least one selected recommendation in the message.

(12) Another object of the embodiments herein is to automatically configure at least one relevant electronic in the second IoT network based on the at least one selected recommendation.

SUMMARY

(13) Accordingly the embodiments herein provide a method predicting content based

recommendations. The method includes recognizing, by a first electronic device in a first IoT network, content displayed on a display of the first electronic device. Further, the method includes generating, by the first electronic device, at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic devices in the first IoT network. Further, the method includes displaying, by the first electronic device, the at least one recommendation on a display of the first electronic device.

(14) In an embodiment, the method further includes detecting, by the first electronic device, the at least one recommendation selected by the user. Further, the method includes automatically including, by the first electronic device, the at least one selected recommendation to compose a message. Further, the method includes sending, by the first electronic device, the message to a second IoT network.

(15) In an embodiment, the method further includes receiving, by the first electronic device in the second IoT network, the message including the content and the at least one selected recommendation from the first electronic device in the first IoT network. Further, the method includes determining, by the first electronic device in the second IoT network, at least one electronic device in the second IoT network, relevant to the at least one selected recommendation in the message. Further, the method includes automatically configuring, by the first electronic device in the second IoT network, at least one relevant electronic device in the second IoT network based on the at least one selected recommendation.

(16) In an embodiment, the at least one recommendation is transferrable to other devices.

(17) In an embodiment, generating, by the first electronic device, the at least one recommendation corresponding to the content, includes extracting, by the first electronic device, actionable data items from the displayed content, determining, by the first electronic device, the at least one electronic device in the first IoT network, relevant to the actionable data items, generating, by the first electronic device, the at least one recommendation representing the displayed content by fetching actions related to at least one relevant electronic device in the first IoT network based on a timestamp of a user activity and at least one electronic device on which the user activity is performed.

(18) In an embodiment, the actionable data items are displayed on the first electronic device in one of a different color, an icon and a tagged user interface.

(19) In an embodiment, the at least one recommendation indicates an information of at least one user activity performed on the at least one electronic device in the first IoT network and a control configuration for at least one electronic device in a second IoT network to perform the at least one user activity.

(20) Accordingly the embodiments herein provide a system for predicting content based recommendations. The system includes a first IoT network and a second IoT network. A first electronic device in the first IoT network is configured to recognize content displayed on a display of the first electronic device. The first electronic device is configured to generating at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic devices in the first IoT network. The first electronic device is configured to displaying the at least one recommendation on a display of the first electronic device. For example, the first electronic device is configured to provide the at least one recommendation on an action prediction panel of a keyboard of the first electronic device. The first electronic device is configured to detect the at least one recommendation selected by the user. The first electronic device is configured to automatically include the at least one selected recommendation to compose a message. The first electronic device is configured to send the message to a second IoT network. A first electronic device in the second IoT network is configured to receive the message including the content and the at least one selected recommendation from the first electronic device. The first electronic device in the second IoT

network is configured to determine at least one electronic device in the second IoT network, relevant to the at least one selected recommendation in the message. The first electronic device in the second IoT network is configured to automatically configure at least one relevant electronic device based on the at least one selected recommendation.

(21) These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

(22) According to various embodiments of the present disclosure, a method and system for predicting content based recommendation may be provided.

(23) According to various embodiments of the present disclosure, at least one recommendation corresponding to the content may be predicted based on a history of user activities performed by the first user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This method and system is illustrated in the accompanying drawings, throughout which like reference letters indicate corresponding parts in the various figures. The embodiments herein will be better understood from the following description with reference to the drawings, in which:

(2) FIG. 1 is a block diagram of a system for predicting content based recommendations and configuring at least one relevant electronic device in a second IoT network based on at least one selected recommendation, according to an embodiment as disclosed herein;

(3) FIG. 2 is a flow diagram illustrating a method for predicting the content based recommendations and configuring the at least one relevant electronic device in the second IoT network based on the at least one selected recommendation, according to an embodiment as disclosed herein;

(4) FIG. 3 is a flow diagram illustrating a method for storing a semantic relation between at least one electronic device and activities of a user, according to an embodiment as disclosed herein;

(5) FIG. 4 is a flow diagram illustrating a method for providing the at least one recommendation to the user for a content displayed on a display of the first electronic device in a first IoT network, according to an embodiment as disclosed herein;

(6) FIG. 5 is a flow diagram illustrating a method for displaying the control information on the relevant electronic devices in the second IoT network based on the availability of the corresponding electronic device, according to an embodiment as disclosed herein;

(7) FIGS. 6a, and 6b are a flow diagram illustrating a method for configuring the corresponding electronic device in the second IoT network based on the at least one selected recommendation, according to an embodiment as disclosed herein;

(8) FIGS. 7a and 7b illustrate an example scenario of predicting the content based recommendations and configuring a smart TV in the second IoT network to play a video based on a selected recommendation, according to an embodiment as disclosed herein;

(9) FIGS. 8a, 8b and 8c are a sequential diagram illustrating signaling in the system for predicting the content based recommendations and configuring the smart TV in the second IoT network to play the video based on the selected recommendation, according to an embodiment as disclosed herein;

(10) FIGS. 9a and 9b illustrate an example scenario of predicting the content based

recommendations and configuring a microwave oven in the second IoT network to bake a cake based on the selected recommendation, according to an embodiment as disclosed herein;

(11) FIG. 10 is a flow diagram illustrating a method for providing the recommendation of the video of baking the cake watched in the smart TV to the user of a smartphone in the first IoT network, according to an embodiment as disclosed herein; and

(12) FIGS. 11a and 11b illustrate an example scenario of predicting the content based recommendations and configuring a smartphone in the second IoT network to play the video based on the selected recommendation, according to an embodiment as disclosed herein.

DETAILED DESCRIPTION

(13) The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. Also, the various embodiments described herein are not necessarily mutually exclusive, as some embodiments can be combined with one or more other embodiments to form new embodiments. The term “or” as used herein, refers to a non-exclusive or, unless otherwise indicated. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein can be practiced and to further enable those skilled in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

(14) As is traditional in the field, embodiments may be described and illustrated in terms of blocks which carry out a described function or functions. These blocks, which may be referred to herein as managers, units, modules, hardware components or the like, are physically implemented by analog and/or digital circuits such as logic gates, integrated circuits, microprocessors, microcontrollers, memory circuits, passive electronic components, active electronic components, optical components, hardwired circuits and the like, and may optionally be driven by firmware and software. The circuits may, for example, be embodied in one or more semiconductor chips, or on substrate supports such as printed circuit boards and the like. The circuits constituting a block may be implemented by dedicated hardware, or by a processor (e.g. one or more programmed microprocessors and associated circuitry), or by a combination of dedicated hardware to perform some functions of the block and a processor to perform other functions of the block. Each block of the embodiments may be physically separated into two or more interacting and discrete blocks without departing from the scope of the disclosure. Likewise, the blocks of the embodiments may be physically combined into more complex blocks without departing from the scope of the disclosure.

(15) Accordingly the embodiments herein provide a system for predicting content based recommendations. The system includes a first IoT network and a second IoT network. A first electronic device in the first IoT network is configured to recognize content displayed on a display of the first electronic device. The first electronic device is configured to predict at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic devices in the first IoT network. For example, the first electronic device is configured to generate at least one recommendation corresponding to the content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic devices in the first IoT network. The first electronic device is configured to display the at least one recommendation on a display of the first electronic device. For example, the first electronic device is configured to provide the at least one recommendation on an action prediction panel of a keyboard of the first electronic device. The first electronic device is configured to detect the at least one recommendation selected by the user. The first electronic device is configured to automatically include the at least one selected recommendation to compose a message. The first electronic device

is configured to send the message to a second IoT network. A first electronic device in the second IoT network is configured to receive the message including the content and the at least one selected recommendation from the first electronic device. The first electronic device in the second IoT network is configured to determine at least one electronic device in the second IoT network, relevant to the at least one selected recommendation in the message. The first electronic device in the second IoT network is configured to automatically configure at least one relevant electronic device based on the at least one selected recommendation.

(16) Unlike to the existing methods and systems, the first electronic device generates a semantic relation of the user activities to the associated electronic devices. Further, the first electronic device classifies the user activities which related to same event to a group, where examples for the event is baking a cake, body fitness etc. The first electronic device determines and stores a control configuration of each associated electronic device while performing the user activities. In response to detecting a chat message displayed in the display, the first electronic device generates the recommendations include the control configuration based on a chat message. The first electronic device generates the recommendations based on a relation between the chat message and the event. Therefore, the control configuration in each recommendation is relevant to the chat message. The user can share at least one recommendation as an action tag (e.g. a tagged text) to other electronic devices by selecting the at least one recommendation. Further, the user can apply the control configuration in the other electronic devices to perform the user activity by selecting the action tag. Therefore, the proposed method improves the user experience in sharing the control configuration to other electronic devices.

(17) Referring now to the drawings, and more particularly to FIGS. **1** through **11**, there are shown preferred embodiments.

(18) FIG. **1** is a block diagram of a system **1000** for predicting content based recommendations and configuring at least one relevant electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in a second IoT network **200** based on at least one selected recommendation, according to an embodiment as disclosed herein. In an embodiment, the system **1000** includes a first IoT network **100** and at least one second IoT network **200**, where the first IoT network **100** is connected to the at least one second IoT network **200**. The first IoT network **100** includes at least one electronic device (i.e. first electronic device **100A-n.sup.th** electronic device **100F**). A second IoT network **200** includes at least one electronic device (i.e. first electronic device **200A-n.sup.th** electronic device **200F**). Examples for the at least one electronic device (i.e. first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** and the at least one electronic device (i.e. first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** are, but not limited to a smartphone, a tablet computer, a personal computer, a desktop computer, a smart watch, a personal digital assistant (PDA), a multimedia device, an Internet of Things (IoT) or the like.

(19) In an embodiment, the first electronic device **100A** includes a processor **110A**, a memory **120A**, a display **130A** and a communicator **140A**, where the processor **110A** is coupled to the memory **120A** and the display **130A**. In an embodiment, the processor **110A** includes a content recognizer **111A**, a recommendation engine **112A**, a device configuration engine **113A**, a semantic relation generator **114A**, and a Machine Learning (ML) engine **115A**. The processor **110A** is configured to execute instructions stored in the memory **120A**. The processor **110A** is configured to detect a keyboard invocation. In an embodiment, the memory **120A** includes a semantic instance relations database **121A**. The memory **120A** may include non-volatile storage elements. Examples of such non-volatile storage elements may include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of an Electrically Programmable Memory (EPROM) or an Electrically Erasable and Programmable Memory (EEPROM).

(20) In addition, the memory **120A** may, in some examples, be considered a non-transitory storage medium. The term “non-transitory” may indicate that the storage medium is not embodied in a

carrier wave or a propagated signal. However, the term “non-transitory” should not be interpreted that the memory **120A** is non-movable. In some examples, the memory **120A** can be configured to store larger amounts of information than the memory **120A**. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g. in Random Access Memory (RAM) or cache).

(21) In an embodiment, the display **130A** is a touch screen display. Examples for the display **130A** are, but not limited to a Liquid Crystal Display (LCD) screen, a Light Emitting Diode (LED) display screen, and the like.

(22) The communicator **140A** is configured to communicate internally between hardware components in the first electronic device **100A**. Further, the communicator **140A** is configured to facilitate the communication between the electronic device **100A** and the second electronic device **100B-n.sup.th** electronic device **100F**. Further, the communicator **140A** is configured to facilitate the communication between the electronic device **100A** and the second IoT network **200**.

(23) In an embodiment, the second electronic device **100A-n.sup.th** electronic device **100F** includes similar hardware components of the first electronic device **100A**. The hardware components in the second electronic device **100A-n.sup.th** electronic device **100F** performs same operations perform by the similar hardware component in the first electronic device **100A**. In another embodiment, the first electronic device **100A** operates as an IoT network server of the first IoT network **100**.

(24) In an embodiment, the first electronic device **200A** includes a processor **210A**, a memory **220A**, a display **230A** and a communicator **240A**, where the processor **210A** is coupled to the memory **220A** and the display **230A**. In an embodiment, the processor **210A** includes a recommendation engine **211A** and a device configuration engine **212A**. The processor **210A** is configured to execute instructions stored in the memory **220A**. The processor **210A** is configured to detect the keyboard invocation. The memory **220A** may include non-volatile storage elements. Examples of such non-volatile storage elements may include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of an Electrically Programmable Memory (EPROM) or an Electrically Erasable and Programmable Memory (EEPROM).

(25) In addition, the memory **220A** may, in some examples, be considered a non-transitory storage medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. However, the term “non-transitory” should not be interpreted that the memory **220A** is non-movable. In some examples, the memory **220A** can be configured to store larger amounts of information than the memory **220A**. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g. in Random Access Memory (RAM) or cache).

(26) The communicator **240A** is configured to communicate internally between hardware components in the first electronic device **200A**. Further, the communicator **240A** is configured to facilitate the communication between the electronic device **200A** and the second electronic device **200B-n.sup.th** electronic device **200F**. Further, the communicator **140A** is configured to facilitate the communication between the electronic device **200A** and the first IoT network **100**.

(27) In an embodiment, the display **230A** is the touch screen display. Examples for the display **230A** are, but not limited to a LCD screen, a LED display screen, and the like.

(28) In an embodiment, the second electronic device **200A-n.sup.th** electronic device **200F** includes similar hardware components of the first electronic device **200A**. The hardware components in the second electronic device **100A-n.sup.th** electronic device **100F** performs same operations perform by the similar hardware component of the first electronic device **200A**. In another embodiment, the first electronic device **200A** operates as the IoT network server of the second IoT network **200**.

(29) The first electronic device **100A** is configured to detect user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. Examples for the user activities are, but not limited to, reading a description about Zumba dance in a webpage, watching a video of Yoga poses, baking a cake, doing workout,

doing work out on treadmill at 10 km/h for 10 minutes, and the like. In an embodiment, the semantic relation generator **114A** detects the user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**.

(30) The first electronic device **100A** is configured to classify the user activities related to same event to a group (e.g. a first group represents activities related to body fitness, a second group represents activities related to baking the cake, a third group represents watching a match, etc.), in response to detecting the user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**). Examples for the event are, but not limited to, the body fitness, baking the cake, watching the match. In an embodiment, the user activities are classified along with additional information of each user activity. In an embodiment, the additional information includes the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) on which the user activity is performed, an application and link used to perform the user activity in the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**), a timestamp of each user activity, a device configuration of each electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) to perform the user activity and a metadata associated with each user activity.

(31) In an embodiment, the semantic relation generator **114A** classifies the user activities related to the same event to the group, in response to detecting the user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**). In an embodiment, the semantic relation generator **114A** determines data includes an information of the application used to perform the user activity, objects identified using Optical Character Recognition (OCR) technique, image recognition technique and speech to text conversion technique and a context of a user of first electronic device **100A**.

(32) Examples for the context of the user of first electronic device **100A** are, but not limited to a time at which the user is performing the user activity, a location of the user of first electronic device **100A**, an occasion at which the user of first electronic device **100A** performs the user activity, etc. Further, the semantic relation generator **114A** filters the data using available knowledge databases (e.g. wikiHow®, Reddit®) and extracts the objects which is frequently identified. Further, the semantic relation generator **114A** determines a relation of the user activities with the event based on the extracted objects.

(33) The first electronic device **100A** is configured to record the classified user activities along with the additional information of each user activity. In an embodiment, the semantic relation generator **114A** records the classified user activities along with the additional information of each user activity to the memory **120A**.

(34) The first electronic device **100A** is configured to define a meaning to each user activity performed on various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at various timestamp. In an embodiment, the metadata is used to define the meaning to each user activity performed on various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at various timestamp. In an embodiment, the ML engine **115A** defines the meaning of each user activity performed on the various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at the various timestamp. In an embodiment, the meaning is defined for each user activity performed on various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at various timestamp indicates a semantic relation of each user activity to the event. In an embodiment, the semantic relation generator **114A** generates the semantic relation of each user activity to the event.

(35) The first electronic device **100A** is configured to convert each record to an action based on the semantic relation. Example for the actions include, but not limited to showing the video in a smart TV, operating the treadmill, operate a microwave oven to bake the cake. In an embodiment, the recommendation engine **112A** converts each record to the action based on the meaning.

(36) The first electronic device **100A** is configured to attach an action tag with the associated

electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**, where the action tag is used to control device settings of the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**. In an embodiment, the recommendation engine **112A** attaches the action tag with the associated electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**.

(37) The first electronic device **100A** is configured to store the action and the action tag to the memory **120A**. In an embodiment, the semantic instance relations database **121A** stores the action and the action tag. The semantic instance relations database **121A** stores hot words related to an actionable data item in the content. In an example scenario, “fit” is the actionable data item in the content. The first electronic device **100A** detects the hot words such as Zumba, Yoga, diet plan, exercise, weight loss related to the actionable data item “fit”, which stores in the semantic instance relations database **121A**.

(38) The first electronic device **100A** is configured to recognize the content displayed on the display **130A** of the first electronic device **100A**. An example for the content is a text, an image, a video, an audio file, a sticker and the like. In an embodiment, the content is recognized by identifying the object (e.g. noun, verb, cake, Zumba dance in the image, Yoga poses in the video, etc.) in the content. In an example, the first electronic device **100A** identify the objects in the content by analyzing the content using an Optical Character Recognition (OCR) technique, image recognition technique, speech to text conversion technique etc. In an embodiment, the content recognizer **111A** recognizes the content displayed on the display **130A**.

(39) The first electronic device **100A** is configured to determine a list of electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) relevant to the hot words. In an embodiment, the recommendation engine **112A** determines the list of electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) relevant to the hot words.

(40) The first electronic device **100A** is configured to generate at least one recommendation corresponding to the content based on a history of the user activities performed using at least one of the first electronic device **100A** and other electronic devices. The first electronic device **100A** is configured to predict at least one recommendation corresponding to the content based on a history of the user activities performed using at least one of the first electronic device **100A** and other electronic devices (second electronic device **100B-n.sup.th** electronic device **100F**) in the first IoT network **100**. In an embodiment, the at least one recommendation indicates an information of at least one user activity performed on the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** and a control configuration (i.e. IoT settings) for the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** to perform the at least one user activity.

(41) In an embodiment, the at least one recommendation is transferrable to the electronic devices (second electronic device **100B-n.sup.th** electronic device **100F**) in the first IoT network and the electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**.

(42) In an embodiment, the recommendation engine **112A** predicts the at least one recommendation corresponding to the content based on the history of the user activities performed by the user using at least one of the first electronic device **100A** and other electronic devices (second electronic device **100B-n.sup.th** electronic device **100F**) in the first IoT network **100**. For example, the recommendation engine **112A** generates the at least one recommendation corresponding to the content based on the history of the user activities performed by the user using at least one of the first electronic device **100A** and other electronic devices in the first IoT network. Further, the recommendation engine **112A** improvise the control configuration in the at least one recommendation by detailing the control configuration. In an example scenario, the control configuration of the microwave oven for baking the cake is 120° C. for 10 minutes. In response to

improvising the control configuration, the control configuration is modified to: switch on the microwave when the voltage is in between a range of 200V-220V, set a temperature of an air conditioner to 28° C., set a temperature of microwave oven to 120° C. for 10 minutes, when a door of the microwave oven is closed.

(43) In another embodiment, the first electronic device **100A** is configured to extract actionable data items from the displayed content. In an embodiment, the actionable data items are displayed on the first electronic device **100A** in one of a different color, an icon and a tagged User Interface (UI). In an embodiment, the content recognizer **111A** extracts the actionable data items from the displayed content. The first electronic device **100A** is configured to determine the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**), relevant to the actionable data items.

(44) In an embodiment, the ML engine **115A** determines the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**), relevant to the actionable data items. The first electronic device **100A** is configured to short list and sort the relevant electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in an order by referring to the records in the memory **120A**. In an embodiment, the recommendation engine **112A** short lists and sorts the relevant electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the order by referring to the records in the memory **120A**.

(45) The first electronic device **100A** is configured to predict the at least one recommendation representing the displayed content by fetching actions related to at least one relevant electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** based on a timestamp of the user activity and at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) on which the user activity is performed. In an embodiment, the first electronic device **100A** is configured to generate the at least one recommendation representing the displayed content by fetching actions related to at least one relevant electronic device in the first IoT network **100** based on a timestamp of the user activity and at least one electronic device on which the user activity is performed. In an embodiment, the recommendation engine **112A** predicts (or generate) the at least one recommendation representing the displayed content by fetching the actions related to the at least one relevant electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** based on the timestamp of the user activity and at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) on which the user activity is performed. The actions related to the electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) are predicted in the sorted order in the list by the recommendation engine **112A** for fetching the actions in the sorted order.

(46) The first electronic device **100A** is configured to display the at least one recommendation on a display of the first electronic device. In an embodiment, the first electronic device **100A** is configured to provide the at least one recommendation on an action prediction panel (not shown) of a keyboard (not shown) of the first electronic device **100A**, where the keyboard displays on the display **130A**. In an example, the action prediction panel is a space in the keyboard located above uppermost keys in the keyboard. In an embodiment, the recommendation displays as a representation of at least one of the user activity, the electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) associated with the user activity, the application associated with the user activity. In an embodiment, the recommendation engine **112A** displays the at least one recommendation on a display **130A** of the first electronic device **100A**. In an embodiment, the recommendation engine **112A** provides the at least one recommendation on the action prediction panel of the keyboard of the first electronic device **100A**. In an embodiment, predictions overlays on the action prediction panel of the keyboard of the first electronic device **100A** for providing the at least one recommendation to the user.

(47) The first electronic device **100A** is configured to detect the at least one recommendation

selected by the user. In an example, the user taps on the at least one recommendation displayed on the display **130A**, for selecting at least one recommendation. In an embodiment, the recommendation engine **112A** detects the at least one recommendation selected by the user. The first electronic device **100A** is configured to automatically include the at least one selected recommendation to compose a message. Examples for the message are a text message, a voice call, a video call, a voice message, an image, a video message, a sticker and the like. In an embodiment, the device configuration engine **113A** automatically includes the at least one selected recommendation to compose the message. In an embodiment, the at least one selected recommendation is included in the message by attaching the action tag to the message.

(48) The first electronic device **100A** is configured to send the message to the second IoT network **200**. In an embodiment, the device configuration engine **113A** sends the message to the second IoT network **200**.

(49) The second electronic device **200A** is configured to receive the message including the content and the at least one selected recommendation from the first electronic device **100A**. In an embodiment, the recommendation engine **211A** receives the message including the content and the at least one selected recommendation from the first electronic device **100A**. The second electronic device **200A** is configured to identify the selected recommendation with which the electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** are tagged. The recommendation engine **211A** identifies the selected recommendation with which the electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** are tagged.

(50) The second electronic device **200A** is configured to determine the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**, relevant to the at least one selected recommendation in the message. In an embodiment, the device configuration engine **212A** determines the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**, relevant to the at least one selected recommendation in the message. The first electronic device **200A** is configured to automatically configure the at least one relevant electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) based on the at least one selected recommendation. In an embodiment, the device configuration engine **212A** automatically configures the at least one relevant electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) based on the at least one selected recommendation.

(51) In another embodiment, the device configuration engine **212A** determines whether the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **100** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the second IoT network **200**, in response to receiving the message from the first electronic device **100A** by the first electronic device **200A**. The device configuration engine **212A** automatically configures the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) available in the second IoT network **200** based on the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **100** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the second IoT network **200**.

(52) The device configuration engine **212A** determines at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** which is relevant to the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is unavailable in the second IoT network **100** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the second IoT network **200**. Further, the device configuration engine **212A** is configured to display the recommendation on the at least one relevant electronic device

(first electronic device 200A-n.sup.th electronic device 200F) in the second IoT network 200.

(53) In another embodiment, the device configuration engine 212A analyzes the control configuration for the electronic devices (first electronic device 100A-n.sup.th electronic device 100F) in second IoT network 200, in response to receiving the message from the first electronic device 100A by the first electronic device 200A. Further, the device configuration engine 212A determines whether the corresponding electronic device (first electronic device 200A-n.sup.th electronic device 200F) is available in second IoT network 200. The device configuration engine 212A applies the control configuration on the corresponding electronic devices (first electronic device 200A-n.sup.th electronic device 200F), when the corresponding electronic device (first electronic device 200A-n.sup.th electronic device 200F) is available in second IoT network 200. The device configuration engine 212A determine and sort nearby relevant electronic devices (first electronic device 200A-n.sup.th electronic device 200F), when the corresponding electronic device (first electronic device 200A-n.sup.th electronic device 200F) is unavailable in second IoT network 200. Further, the device configuration engine 212A is configured to display the control configuration as a text on the display (not shown) of the relevant electronic devices (first electronic device 200A-n.sup.th electronic device 200F).

(54) In another embodiment, the device configuration engine 212A determines whether the at least one electronic device (first electronic device 200A-n.sup.th electronic device 200F) is available for a user of the second electronic device 200A, in response to receiving the message from the first electronic device 100A by the first electronic device 200A. Further, the device configuration engine 212A deletes the message, when the at least one electronic device (first electronic device 200A-n.sup.th electronic device 200F) is unavailable for the user of the second electronic device 200A. The device configuration engine 212A queries the list of electronic device (first electronic device 200A-n.sup.th electronic device 200F) available for the user to the IoT network server in the second IoT network 200, when the at least one electronic device (first electronic device 200A-n.sup.th electronic device 200F) is available for the user of the second electronic device 200A.

(55) The device configuration engine 212A determines whether the electronic device (first electronic device 200A-n.sup.th electronic device 200F) is compatible to Open Connectivity Foundation (OCF) or other IoT compatibility standards. The device configuration engine 212A queries device properties of the similar electronic device (first electronic device 200A-n.sup.th electronic device 200F), when the electronic device (first electronic device 200A-n.sup.th electronic device 200F) is compatible to the OCF or the other IoT compatibility standards. Examples for the device properties are, but not limited to a capability of the device to play the video, a capability of the device to play an audio, a capability of the device to display a text, etc.

(56) The device configuration engine 212A determines whether the electronic device (first electronic device 200A-n.sup.th electronic device 200F) supports by proprietary technologies/certifications (e.g. Work With SmartThings (WWST), works with Alexa, Compatible with SmartThings (CST), etc.), when the electronic device (first electronic device 200A-n.sup.th electronic device 200F) is incompatible to the OCF or the other IoT compatibility standards. The device configuration engine 212A performs non-standard conversion of control configuration based on the first electronic device 200A in the second IoT network 200, when the electronic device (first electronic device 200A-n.sup.th electronic device 200F) does not support by the proprietary technologies/certifications.

(57) Further, the device configuration engine 212A generates the list of the electronic devices (first electronic device 200A-n.sup.th electronic device 200F) where the first electronic device 200A have the access. The device configuration engine 212A queries the device properties of the similar electronic device (first electronic device 200A-n.sup.th electronic device 200F), when the electronic device (first electronic device 200A-n.sup.th electronic device 200F) supports by the proprietary technologies/certifications. Further, the device configuration engine 212A determines whether the device properties of the first electronic device 200A in the second IoT network 200 are

same to the device properties of first electronic device **100A** in the first IoT network **100**. The device configuration engine **212A** determines base properties of both the electronic devices (i.e. first electronic device **100A** and first electronic device **200A**) and attach the base properties, when the device properties of first electronic device **200A** in the second IoT network **200** are not same to the device properties of the first electronic device **100A** in the first IoT network **100**. The base properties are a set of device properties which are available common in both the first electronic device **100A** and the first electronic device **200A**.

(58) The device configuration engine **212A** attaches the device properties, when the device properties of the first electronic device **200A** in the second IoT network **200** are same to the device properties of first electronic device **100A** in the first IoT network **100**. In response to attaching the device properties or the base properties, the device configuration engine **212A** generates the list of the electronic devices (first electronic device **200A**-n.sup.th electronic device **200F**) where the first electronic device **200A** have the access. Further, the device configuration engine **212A** sorts the electronic devices (first electronic device **200A**-n.sup.th electronic device **200F**) based on a preference and a context of the user of the first electronic device **200A**.

(59) In an example scenario, the user of first electronic device **200A** prefers to watch the video in the smart TV apart from watching the video in the smartphone. Examples for the context of the user of first electronic device **200A** include, but not limited to a time at which the user of the first electronic device **200A** views the message, a location of the user of the first electronic device **200A**, an occasion at which the user of the first electronic device **200A** receives the message, etc. The device configuration engine **212A** selects the recommendation based on the user selection. Further, the device configuration engine **212A** applies the control configuration in the corresponding electronic device (first electronic device **200A**-n.sup.th electronic device **200F**).

(60) Although the FIG. **1** shows the hardware components of the electronic device **100** but it is to be understood that other embodiments are not limited thereon. In other embodiments, the electronic device **100** may include less or more number of components. Further, the labels or names of the components are used only for illustrative purpose and does not limit the scope of the invention. One or more components can be combined together to perform same or substantially similar function for predicting content based recommendations and configuring at least one relevant electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) in the second IoT network **200** based on the at least one selected recommendation.

(61) FIG. **2** is a flow diagram **2000** illustrating a method for predicting the content based recommendations and configuring the at least one relevant electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) in the second IoT network **200** based on the at least one selected recommendation, according to an embodiment as disclosed herein. At **201**, the method includes recognizing the content displayed on the display **130A** of the first electronic device **100A**. In an embodiment, the method allows the content recognizer **111A** to recognize the content displayed on the display **130A** of the first electronic device **100A**. At **202**, the method includes predicting the at least one recommendation corresponding to the content based on the history of the user activities performed by the user using at least one of the first electronic device **100A** and other electronic devices (second electronic device **100B**-n.sup.th electronic device **100F**) in the first IoT network **100**. In an embodiment, the method includes generating the at least one recommendation corresponding to the content based on the history of the user activities performed by the user using at least one of the first electronic device **100A** and other electronic devices (second electronic device **100B**-n.sup.th electronic device **100F**) in the first IoT network **100**. In an embodiment, the method allows the recommendation engine **112A** to predict the at least one recommendation corresponding to the content based on the history of the user activities performed by the user using at least one of the first electronic device **100A** and other electronic devices (second electronic device **100B**-n.sup.th electronic device **100F**) in the first IoT network **100**.

(62) At **203**, the method includes providing the at least one recommendation on the action

prediction panel of the keyboard of the first electronic device **100A**. In an embodiment, the method includes displaying the at least one recommendation on a display **130A** of the first electronic device **100A**. In an embodiment, the method allows the recommendation engine **112A** to provide the at least one recommendation on the action prediction panel of the keyboard of the first electronic device **100A**. At **204**, the method includes detecting the at least one recommendation selected by the user. In an embodiment, the method allows the recommendation engine **112A** to detect the at least one recommendation selected by the user. At **205**, the method includes automatically including the at least one selected recommendation to compose the message. In an embodiment, the method allows the device configuration engine **113A** to automatically include the at least one selected recommendation to compose the message. At **206**, the method includes sending the message to the second IoT network **200**. In an embodiment, the method allows the device configuration engine **113A** to send the message to the second IoT network **200**. At **207**, the method includes receiving the message including the content and the at least one selected recommendation from the first electronic device **100A**. In an embodiment, the method allows the recommendation engine **211A** to receive the message including the content and the at least one selected recommendation from the first electronic device **100A**.

(63) At **208**, the method includes determining whether the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. In an embodiment, the method allows the device configuration engine **212A** to determine whether the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. At **209**, the method includes automatically configuring the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) available in the second IoT network **200** based on the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. In an embodiment, the method allows the device configuration engine **212A** to automatically configure the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) available in the second IoT network **200** based on the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**.

(64) At **210**, the method includes determining the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** which is relevant to the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is unavailable in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. In an embodiment, the method allows the device configuration engine **212A** to determines the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** which is relevant to the at least one selected recommendation, when the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is unavailable in the second IoT network **200** corresponds to the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. At **211**, the method includes displaying the recommendation on the at least one relevant electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**. In an embodiment, the method allows the device configuration engine **212A** to display the recommendation on the at least one relevant electronic device (first

electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**.

(65) The various actions, acts, blocks, steps, or the like in the flow diagram **2000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(66) FIG. **3** is a flow diagram **3000** illustrating steps in storing the semantic relation between at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) and activities of the user, according to an embodiment as disclosed herein. At **301**, the method includes detecting the user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. In an embodiment, the method allows the semantic relation generator **114A** to detect the user activities associated with the at least one electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**. At **302**, the method includes classifying the activities related to the same event to the group. In an embodiment, the method allows the semantic relation generator **114A** to classify the activities related to the same event to the group. At **303**, the method includes recording the classified user activities along with the additional information of each user activity. In an embodiment, the method allows the semantic relation generator **114A** to record the classified user activities along with the additional information of each user activity. At **304**, the method includes defining the meaning to each user activity performed on various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at various timestamp. In an embodiment, the method allows the ML engine **115A** to define the meaning to each user activity performed on various electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) at various timestamp.

(67) At **305**, the method includes converting each record to the action based on the meaning. In an embodiment, the method allows the recommendation engine **112A** to convert each record to the action based on the meaning. At **306**, the method includes attaching the action tag with the associated electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**, where the action tag is used to control the device settings of the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**. In an embodiment, the method allows the recommendation engine **112A** to attach the action tag with the associated electronic device (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100**, where the action tag is used to control the device settings of the at least one electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**. At **307**, the method includes storing the action and the action tag. In an embodiment, the method allows the semantic instance relations database **121A** to store the action and the action tag.

(68) The various actions, acts, blocks, steps, or the like in the flow diagram **3000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(69) FIG. **4** is a flow diagram **4000** illustrating steps in providing the at least one recommendation to the user for the content displayed on the display **130** of the first electronic device **100A** in the first IoT network **100**, according to an embodiment as disclosed herein. At **401**, the method includes analyzing the content using the OCR and image recognition techniques for determining the actionable data items related to the event. In an embodiment, the method allows the content recognizer **111A** to analyze the content using the OCR and image recognition techniques for determining the actionable data items related to the event. At **402**, the method includes identifying the electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** which are relevant to the actionable data items in the content. In an embodiment, the method allows the recommendation engine **112A** to identify the electronic devices (first electronic

device **100A-n.sup.th** electronic device **100F**) in the first IoT network **100** which are relevant to the actionable data items in the content.

(70) At **403**, the method includes short listing and sorting the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) by referring to the records in the memory **120A**. In an embodiment, the method allows the recommendation engine **112A** to short list and sort the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) by referring to the records in the memory **120A**. At **404**, the method includes predicting the actions associated with the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the sorted order in the list. In an embodiment, the method includes generating the actions associated with the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the sorted order in the list. In an embodiment, the method allows the recommendation engine **112A** to predict the actions associated with the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the sorted order in the list. In an embodiment, the method allows the recommendation engine **112A** to generate the actions associated with the relevant electronic devices (first electronic device **100A-n.sup.th** electronic device **100F**) in the sorted order in the list. At **405**, the method includes overlaying the recommendations on the display of the first electronic device. In embodiment, the method includes overlaying the recommendations on the action prediction panel of the keyboard of the first electronic device **100A**. In an embodiment, the method allows the recommendation engine **112A** to overlay the recommendations on the action prediction panel of the keyboard of the first electronic device **100A**.

(71) At **406**, the method includes creating the message with attached action tag includes the control configuration based on the user selection on the recommendation. In an embodiment, the method allows the device configuration engine **113A** to create the message with attached action tag includes the control configuration based on the user selection on the recommendation. At **407**, the method includes sending the message to the second IoT network **200**. In an embodiment, the method allows the device configuration engine **113A** to send the message to the second IoT network **200**.

(72) The various actions, acts, blocks, steps, or the like in the flow diagram **4000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(73) FIG. 5 is a flow diagram **5000** illustrating steps in displaying the control information on the relevant electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200** based on the availability of the corresponding electronic device (first electronic device **200A-n.sup.th** electronic device **200F**), according to an embodiment as disclosed herein. At **501**, the method includes analyzing the control configuration for the electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**. In an embodiment, the method allows the device configuration engine **212A** to analyze the control configuration for the electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) in the second IoT network **200**.

(74) At **502**, the method includes determining whether the corresponding electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** based on the control configuration. In an embodiment, the method allows the device configuration engine **212A** to determine whether the corresponding electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200** based on the control configuration. At **503**, the method includes applying the control configuration on the corresponding electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**), when the corresponding electronic device (first electronic device **200A-n.sup.th** electronic device **200F**) is available in the second IoT network **200**. In an embodiment, the method allows the device

configuration engine **212A** to apply the control configuration on the corresponding electronic devices (first electronic device **200A**-n.sup.th electronic device **200F**), when the corresponding electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is available in the second IoT network **200**.

(75) At **504**, the method includes determining and sorting the relevant devices, when the corresponding electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is unavailable in the second IoT network **200**. In an embodiment, the method allows the device configuration engine **212A** to determine and sort the relevant devices, when the corresponding electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is unavailable in the second IoT network **200**. At **502**, the method includes displaying the control information on the relevant electronic devices (first electronic device **200A**-n.sup.th electronic device **200F**). In an embodiment, the method allows the device configuration engine **212A** to display the control information on the relevant electronic devices (first electronic device **200A**-n.sup.th electronic device **200F**).

(76) The various actions, acts, blocks, steps, or the like in the flow diagram **5000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(77) FIGS. **6a** and **6b** are a flow diagram **6000** illustrating steps in configuring the corresponding electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) in the second IoT network **200** based on the at least one selected recommendation, according to an embodiment as disclosed herein. At **601**, the method includes receiving the message includes the content and the selected recommendation. At **602**, the method includes identifying the selected recommendation with which the electronic devices (first electronic device **100A**-n.sup.th electronic device **100F**) in the first IoT network **100** are tagged, where the selected recommendation shows in the different color or tagged UI. At **603**, the method includes determining whether the at least one electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is available for the user of the second electronic device **200A**.

(78) At **604**, the method includes deleting the message, when the at least one electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is unavailable for the user of the second electronic device **200A**. At **605**, the method includes querying the list of electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) available for the user to the IoT network server in the second IoT network **200**, when the at least one electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is available for the user of the second electronic device **200A**. At **606**, the method includes determining whether the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is compatible to the OCF or the other IoT compatibility standards. Further, at **609**, the method includes querying the device properties of the similar electronic device (first electronic device **200A**-n.sup.th electronic device **200F**), when the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is compatible to the OCF or the other IoT compatibility standards. At **607**, the method includes determining whether the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) supports by the proprietary technologies/certifications, when the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) is incompatible to the OCF or the other IoT compatibility standards. The method includes performing step **609**, when the electronic device **100A** the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) supports by the proprietary technologies/certifications.

(79) At **608**, the method includes performing the non-standard conversion of the control configuration based on the first electronic device **200A** in the second IoT network **200**, when the electronic device (first electronic device **200A**-n.sup.th electronic device **200F**) does not support by the proprietary technologies/certifications. Further, at **613**, the method includes generating the list

of the electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) where the first electronic device **200A** have the access. At **610**, the method includes determining whether the device properties of the first electronic device **200A** in the second IoT network **200** are same to the device properties of the first electronic device **100A** in the first IoT network **100**. At **611**, the method includes determining the base properties of both the electronic devices (i.e. first electronic device **100A** and first electronic device **200A**) and attach the base properties, when the device properties of first electronic device **200A** in the second IoT network **200** are not same to the device properties of the first electronic device **100A** in the first IoT network **100**. Further, the method includes performing the step **613**.

(80) At **612**, the method includes attaching the device properties, when the device properties of first electronic device **200A** in the second IoT network **200** are same to the device properties of first electronic device **100A** in the first IoT network **100**. Further, the method includes performing the step **613**. At **614**, the method includes sorting the electronic devices (first electronic device **200A-n.sup.th** electronic device **200F**) based on the preference and the context of the user of first electronic device **200A**. At **615**, the method includes selecting the recommendation based on the user selection. At **616**, the method includes applying the control configuration in the corresponding electronic device (first electronic device **200A-n.sup.th** electronic device **200F**). In an embodiment, the method allows the device configuration engine **212A** to perform from the step **601** to step **616**.

(81) The various actions, acts, blocks, steps, or the like in the flow diagram **6000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(82) FIGS. **7a** and **7b** illustrate an example scenario of predicting the content based recommendations and configuring the smart TV **200B** in the second IoT network **200** to play the video based on the selected recommendation, according to an embodiment as disclosed herein. In the example scenario, a smartphone **100A** is the first electronic device **100A**, a smart TV **100B** is the second electronic device **100B**, a refrigerator **100C** is the third electronic device **100C** and the treadmill **100D** is the fourth electronic device **100D** in the first IoT network **100**. The smartphone **100A** is connected to the smart TV **100B**, the refrigerator **100C** and the treadmill **100D**. In the example scenario, a smartphone **200A** is the first electronic device **200A**, a smart TV **200B** is the second electronic device **200B** in the second IoT network **200**. The smartphone **200A** is connected to the smart TV **200B**. Further, the first IoT network **100** is connected to the second IoT network **200**. Joy is the user of the smartphone **100A**. Jessie the user of the smartphone **200A**. The smartphone **100A** and the smartphone **200A** are supported by the proprietary technologies/certifications.

(83) Consider, that Joy has read a journal on Zumba dance for fit body in the smartphone **100A** at 6 AM of a day as shown in notation (a) of the FIG. **7a**, where the journal is browsed using a browser application in the smartphone **100A**. The smartphone **100A** detects the user activity of reading the journal in the smartphone **100A**. Further, the smartphone **100A** determines that the user activity of reading the journal is related to body fitness, where the body fitness is the event in the example scenario. Further, the smartphone **100A** classifies the user activity of reading the journal to a first group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smartphone **100A** to provide the journal.

(84) Consider, that Joy has seen a video of Yoga poses in the smart TV **100B** at 8 AM of the day as shown in notation (b) of the FIG. **7a**, where the video is played using a video player application in the smart TV **100B**. The smartphone **100A** detects the user activity of watching the video in the smart TV **100B**. Further, the smartphone **100A** determines that the user activity of watching the video is related to the body fitness. Further, the smartphone **100A** classifies the user activity of watching the video to the first group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smart TV **100B** to provide the video.

(85) Consider that Joy has consumed a set of food for the day based on a diet plan from the refrigerator **100C** at 10 AM of the day as shown in notation (c) of the FIG. 7a, where Joy takes different set of food in each day based on the diet plan. The smartphone **100A** detects the user activity of consuming the set of food from the refrigerator **100C**. Further, the smartphone **100A** determines that the user activity of taking the set of food based on the diet plan is related to the body fitness. Further, the smartphone **100A** classifies the user activity of taking the set of food to the first group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the refrigerator **100C** to provide the diet plan.

(86) Consider, that Joy has exercised on the treadmill **100D** at 5 km/h speed for 10 minutes at 5 PM of the day as shown in notation (d) of the FIG. 7a. The smartphone **100A** detects the user activity of exercising on the treadmill **100D**. Further, the smartphone **100A** determines that the user activity of exercising is related to the body fitness. Further, the smartphone **100A** classifies the user activity of exercising to the first group based on the relation. Further, the smartphone **100A** records the control configuration of the treadmill **100D** to operate at 5 km/h speed for 10 minutes.

(87) Consider, Joy receives a text message “Hi, you look so fit in your profile picture. What is the secret?” in the smartphone **100A** from Jessie as shown in notation (e) of the FIG. 7b. Joy inputs as “I followed” in the keyboard for providing a reply to the text message. In response to detecting the user input on the keyboard, the smartphone **100A** detects the object “fit” as the actionable data item. Further, the smartphone **100A** extracts the actionable data item in the text message. Further, the smartphone **100A** determines the electronic devices in the first group as the relevant electronic devices based on the relation of the actionable data item and the user activities classified in the first group. Further, the smartphone **100A** generates four recommendations representing the user activity on each relevant electronic device. In an embodiment, the smartphone **100A** predicts four recommendations representing the user activity on each relevant electronic device. Further, the smartphone **100A** sorts the four recommendations in the order based on the timestamp of each user activity. Further, the smartphone **100A** displays the four recommendations in the sorted order to Joy. In an embodiment, the smartphone **100A** provides the four recommendations in the sorted order to Joy. In response to detecting the user selection on the recommendation represents the user activity of watching the video on the smart TV **100B**, the smartphone **100A** inserts the action tag “video” in the reply message as the icon. The action tag “video” includes the control configuration of the smart TV **100B**.

(88) Consider, Joy sends the reply message to Jessie. In response to receiving the reply message, the smartphone **200A** detects the action tag “video” in the reply message as shown in notation (f) of the FIG. 7b. Further, the smartphone **200A** determines the control configuration associated with the action tag. Further, the smartphone **200A** determines that the smart TV **200B** as the corresponding electronic device to apply the control configuration in the second IoT network **200**. In response to receiving the user selection on the action tag from Jessie, the smartphone **200A** applies the control configuration to the smart TV **200B** to play the video of Yoga poses as shown in notation (g) of the FIG. 7b.

(89) FIGS. 8a, 8b and 8c are a sequential diagram illustrating signaling in the system **1000** for predicting the content based recommendations and configuring the smart TV **200B** in the second IoT network **200** to play the video based on the selected recommendation, according to an embodiment as disclosed herein. The semantic relation generator **114A** determines (801) the user activities associated with the smartphone **100A**-treadmill **100D** performed at various instances. Further, the semantic relation generator **114A** generates (802) the semantic relation of each user activity with the smartphone **100A**-treadmill **100D**. The semantic relation generator **114A** classifies the user activities to the first group based on the relation of the user activity to the event. The device configuration engine **113A** determines the control configuration of the smartphone **100A**-treadmill **100D**. Further, the semantic relation generator **114A** sends (803) the semantic relation to the semantic instance relations database **121A**. The semantic instance relations database **121A**

stores (804) the semantic relation and the control configuration associated with the smartphone **100A**-treadmill **100D**.

(90) Consider, that Joy receives a text message “Hi, you look so fit in your profile picture. What is the secret?” in the smartphone **100A** from the smartphone **200A**. In response to detecting (805) the invocation on the keyboard while Joy provides the input to the smartphone **100A** for sending the reply message “I followed”, the processor **120** sends (806) the displayed content to the content recognizer **111A**. The content recognizer **111A** identifies (807) the object “fit” as the actionable data item in the content. Further, the content recognizer **111A** requests (808) to the semantic instance relations database **121A** to provide the hot words related to the actionable data item. Further, the recommendation engine **112A** fetches (809) the hot words “Zumba, Yoga, diet plan, exercise and weight loss” which are related to the actionable data item from the semantic instance relations database **121A**, in response to sending the request to the semantic instance relations database **121A** by the content recognizer **111A**.

(91) Further, the recommendation engine **112A** determines (810) the list of electronic devices (smartphone **100A**-treadmill **100D**) relevant to the hot words from the first group based on the metadata in the semantic relation. Further, the recommendation engine **112A** generates the recommendations corresponding to each user activity and displays (811) the recommendations to Joy. In an embodiment, the recommendation engine **112A** predicts the recommendations corresponding to each user activity and provides (811) the recommendations to Joy. In response to selecting the recommendation corresponds to the user activity of watching the video in the smart TV **100B**, the recommendation engine **112A** detects (812) the selected recommendation. Further, the recommendation engine **112A** attaches the selected recommendation to the reply message by inserting the action tag “video” to the reply message. The action tag “video” includes the control configuration of the smart TV **100B**. In response to detecting (814) the user selection for sending the reply message, the recommendation engine **112A** sends (815) the reply message to the smartphone **200A**.

(92) In response to receiving the reply message from the smartphone **100A**, the recommendation engine **211A** identifies (815) the selected recommendation in the reply message. Further, the device configuration engine **212A** detects (816) the electronic devices (smartphone **200A**-smart TV **200B**) available for Jessie to use, which is relevant to the recommendation. Further, the device configuration engine **212A** determines (817) the compatibility of the available electronic devices (smartphone **200A**-smart TV **200B**) to apply the control configuration. Further, the device configuration engine **212A** converts the control configuration to a suitable control configuration which is compatible for the available electronic devices (smartphone **200A**-smart TV **200B**), if the control configuration is not compatible to available electronic devices. The device configuration engine **212A** detects that the smart TV **200B** is the corresponding electronic device in the second IoT network **200**, which is available for Jessie. In response to detecting (819) the user selection on the action tag in the reply message by the recommendation engine **211A**, the device configuration engine **212A** apply the control configuration to the smart TV **200B** to play the Yoga video, which is the corresponding electronic device in the second IoT network **200**.

(93) FIGS. **9a** and **9b** illustrate an example scenario of predicting the content based recommendations and configuring a microwave oven **200C** in the second IoT network **200** to bake the cake based on the selected recommendation, according to an embodiment as disclosed herein. In the example scenario, the smartphone **100A** is the first electronic device **100A**, the smart TV **100B** is the second electronic device **100B**, the refrigerator **100C** is the third electronic device **100C** and the microwave oven **100E** is the fourth electronic device **100D** in the first IoT network **100**. The smartphone **100A** is connected to the smart TV **100B**, the refrigerator **100C** and the microwave oven **100E**. In the example scenario, the smartphone **200A** is the first electronic device **200A**, the smart TV **200B** is the second electronic device **200B** and a microwave oven **200C** is the third electronic device **200C** in the second IoT network **200**. The smartphone **200A** is connected to

the smart TV **200B** and the microwave oven **200C**. Further, the first IoT network **100** is connected to the second IoT network **200**. Joy is the user of the smartphone **100A**. Jessie the user of the smartphone **200A**. The smartphone **100A** and the smartphone **200A** are supported by the proprietary technologies/certifications.

(94) Consider, that Joy has seen a video of cherry cake preparation in the smart TV **100B** at 8 AM of the day as shown in notation (a) of the FIG. **9a**, where the video is played using the video player application in the smart TV **100B**. The smartphone **100A** detects the user activity of watching the video in the smart TV **100B**. Further, the smartphone **100A** determines that the user activity of watching the video is related to baking the cake, where baking the cake is the event in the example scenario. Further, the smartphone **100A** classifies the user activity of watching the video to the second group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smart TV **100B** to provide the video.

(95) Consider, that Joy has read a recipe of cherry cake displayed on the display **130C** of the refrigerator **100C** at 10 AM of a day as shown in notation (b) of the FIG. **9a**. The smartphone **100A** detects the user activity of reading the recipe in the refrigerator **100C**. Further, the smartphone **100A** determines that the user activity of reading the recipe is related to baking the cake. Further, the smartphone **100A** classifies the user activity of reading the recipe to the second group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the refrigerator **100C** to provide the recipe.

(96) Consider, that Joy has baked the cherry cake in the microwave oven **100E** at 120° C. for 10 minutes at 5 PM of the day as shown in notation (c) of the FIG. **9a**. The smartphone **100A** detects the user activity of baking the cherry cake using the microwave oven **100E**. Further, the smartphone **100A** determines that the user activity of baking the cherry cake is related to the baking the cake. Further, the smartphone **100A** classifies the user activity of baking the cherry cake to the second group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the microwave oven **100E** to operate at 120° C. for 10 minutes.

(97) Consider, Joy sends an image of cherry cake to Jessie. Further, Joy receives a text message “Hey, it's looking delicious, can you share the recipe” in the smartphone **100A** from Jessie as shown in notation (d) of the FIG. **9a**. Joy inputs as “Yes sure, this is” in the keyboard for providing the reply to the text message. In response to detecting the user input on the keyboard, the smartphone **100A** detects the object “cherry, cake, recipe” as the actionable data items. Further, the smartphone **100A** extracts the actionable data items in the messages displayed in the display **130A**. Further, the smartphone **100A** determines the electronic devices in the second group as the relevant electronic devices based on the relation of the actionable data items and the user activities classified in the second group. Further, the smartphone **100A** generates three recommendations representing the user activity on each relevant electronic device. In an embodiment, the smartphone **100A** predicts three recommendations representing the user activity on each relevant electronic device. Further, the smartphone **100A** sorts the three recommendations in the order based on the timestamp of each user activity. Further, the smartphone **100A** displays the three recommendations in the sorted order to Joy. In an embodiment, the smartphone **100A** provides the three recommendations in the sorted order to Joy. In response to detecting the user selection on the recommendation represents the user activity of baking the cherry cake using the microwave oven **100E**, the smartphone **100A** inserts the action tag “cake's recipe” in the reply message in the different color. The action tag “cake's recipe” includes the control configuration of the microwave oven **100E**.

(98) Consider, Joy sends the reply message to Jessie. In response to receiving the reply message, the smartphone **200A** detects the action tag “cake's recipe” in the reply message as shown in notation (e) of the FIG. **9b**. Further, the smartphone **200A** determines the control configuration associated with the action tag. Further, the smartphone **200A** determines that the microwave oven **200C** as the corresponding electronic device to apply the control configuration in the second IoT

network **200**. In response to receiving the user selection on the action tag from Jessie, the smartphone **200A** applies the control configuration to the microwave oven **200C** to operate at 120° C. for 10 minutes as shown in notation (f) of the FIG. **9b**.

(99) FIG. **10** is a flow diagram **10000** illustrating steps in providing the recommendation of the video of baking the cake watched in the smart TV **100B** to the user of the smartphone **200A** in the first IoT network **100**, according to an embodiment as disclosed herein. At **1001**, the method includes detecting the user activity associated with the smart TV **100**, where the user activity is watching the video of cherry cake preparation. In an embodiment, the method allows the sematic relation generator **114A** to detect the user activity associated with the smart TV **100**. At **1002**, the method includes determining the data related to the user activity. In an embodiment, the method allows the sematic relation generator **114A** to determine the data related to the user activity of watching the video of cherry cake preparation. The data includes application name as video player application, the objects identified using OCR technique are cake and cherry, the object identified using the image recognition technique is cake and the user context (i.e. time: 11:30 AM, Location: Home, Occasion: None).

(100) At **1003**, the method includes filtering the data and extracting the frequent object in the data, where the frequent object is “cake” In an embodiment, the method allows the sematic relation generator **114A** to filter the data and extracting the frequent object in the data. At **1004**, the method includes detecting the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) and the control configuration based on the object. In an embodiment, the method allows the recommendation engine **112A** to detect the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) based on the object. In an embodiment, the method allows the device configuration engine **113A** to detect the control configuration based on the object. At **1005**, the method includes storing the semantic relation includes the details of the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) and the control configuration of the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**). In an embodiment, the method allows the semantic instance relations database **121A** to store the semantic relation includes the details of the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) and the control configuration of the relevant electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**).

(101) At **1006**, the method includes receiving the semantic relation of at least one activity performed on the at least one electronic device (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) in the first IoT network **100** from the sematic instance relation database **121A**, in response to recognizing the content and invoking the keyboard of the smartphone **100A**. In an embodiment, the method allows the recommendation engine **112A** to receive the semantic relation of at least one activity performed on the at least one electronic device (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) in the first IoT network **100** from the sematic instance relation database **121A**, in response to recognizing the content and invoking the keyboard of the smartphone **100A**. At **1007**, the method includes receiving the list of compatible electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) from the video player application. In an embodiment, the method allows the recommendation engine **112A** to receive the list of compatible electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**) from the video player application. At **1008**, the method includes predicting the recommendation for the content displayed in the smartphone **100A** based on the sematic relation and the list of compatible electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**). In an embodiment, the method includes generating the recommendation for the content displayed in the smartphone **100A** based on the sematic relation and the list of compatible electronic devices (smart TV **100B**, refrigerator **100C** and microwave oven **100E**). In an embodiment, the method allows the recommendation engine **112A** to predict the recommendation for the content displayed in the smartphone **100A** based on the sematic relation and the list of compatible electronic devices (smart

TV **100B**, refrigerator **100C** and microwave oven **100E**).

(102) At **1009**, the method includes improvising the control configurations. In an embodiment, the method allows the recommendation engine **112A** to improvise the control configurations. At **1010**, the method includes sorting the recommendations in the order based on the timestamp of the user activity. In an embodiment, the method allows the recommendation engine **112A** to sort the recommendations in the order based on the timestamp of the user activity. At **1011**, the method includes displaying the recommendations on the display of the first electronic device. In an embodiment, the method includes displaying the recommendations on the action prediction panel of the keyboard. In an embodiment, the method allows the device configuration engine **113A** to display the recommendations on the action prediction panel of the keyboard.

(103) The various actions, acts, blocks, steps, or the like in the flow diagram **10000** may be performed in the order presented, in a different order or simultaneously. Further, in some embodiments, some of the actions, acts, blocks, steps, or the like may be omitted, added, modified, skipped, or the like without departing from the scope of the invention.

(104) FIGS. **11a** and **11b** illustrate an example scenario of predicting the content based recommendations and configuring the smartphone **200A** in the second IoT network **200** to play the video based on the selected recommendation, according to an embodiment as disclosed herein. In the example scenario, the smartphone **100A** is the first electronic device **100A** and the smart TV **100B** is the second electronic device **100B** in the first IoT network **100**. The smartphone **100A** is connected to the smart TV **100B**. In the example scenario, the smartphone **200A** is the first electronic device **200A** and the smart TV **200B** is the second electronic device **200B** in the second IoT network **200**, where the smart TV **200B** is unavailable to the user of the smartphone **200A**. The first IoT network **100** is connected to the second IoT network **200**. Joy is the user of the smartphone **100A**. Jessie the user of the smartphone **200A**. The smartphone **100A** and the smartphone **200A** are supported by the proprietary technologies/certifications.

(105) Consider, that Joy has seen a video of football match in the smartphone **100A** at 8 AM of the day as shown in notation (a) of the FIG. **11a**, where the video is played using the video player application in smartphone **100A**. The smartphone **100A** detects the user activity of watching the video in the smartphone **100A**. Further, the smartphone **100A** determines that the user activity of watching the video is related to a match, where the match is the event in the example scenario. Further, the smartphone **100A** classifies the user activity of watching the video to a third group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smartphone **100A** to provide the video.

(106) Consider, that Joy has seen a video of a cricket match in the smart TV **100B** at 10 AM of the day as shown in notation (b) of the FIG. **11a**, where the video is played using the video player application in smart TV **100B**. The smartphone **100A** detects the user activity of watching the video of the cricket match in the smart TV **100B**. Further, the smartphone **100A** determines that the user activity of watching the video of the cricket match is related to the match. Further, the smartphone **100A** classifies the user activity of watching the video of the cricket match to the third group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smart TV **100B** to provide the video of the cricket match.

(107) Consider, that Joy has seen a video of cricket match in the smartphone **100A** at 5 PM of the day as shown in notation (c) of the FIG. **11a**, where the video is played using the video player application in smartphone **100A**. The smartphone **100A** detects the user activity of watching the video of the cricket match in the smartphone **100A**. Further, the smartphone **100A** determines that the user activity of watching the video of the cricket match is related to the match. Further, the smartphone **100A** classifies the user activity of watching the video of the cricket match to the third group based on the relation of the user activity to the event. Further, the smartphone **100A** records the control configuration of the smartphone **100A** to provide the video of the cricket match.

(108) Consider, Joy receives a text message “Hi, Joy. What you done today?” from Jessie as shown

in notation (d) of the FIG. 11a. The user inputs as “Hey Jessie, I watched the match” in the keyboard for providing a reply to the text message. In response to detecting the user input on the keyboard, the smartphone 100A detects the object “watched, match” as the actionable data items. Further, the smartphone 100A extracts the actionable data item in the user input. Further, the smartphone 100A determines the electronic devices in the third group as the relevant electronic devices based on the relation of the actionable data item and the user activities classified in the third group. Further, the smartphone 100A predicts three recommendations representing the user activity on each relevant electronic device. In an embodiment, the smartphone 100A generates three recommendations representing the user activity on each relevant electronic device. Further, the smartphone 100A sorts the three recommendations in the order based on the timestamp of each user activity. Further, the smartphone 100A provides the three recommendations in the sorted order to the user. In an embodiment, the smartphone 100A displays the three recommendations on the display in the sorted order to the user. In response to detecting the user selection on the recommendation represents the user activity of watching the video of the cricket match on the smart TV 100B, the smartphone 100A inserts the tagged user interface to the word “match” as the action tag in the reply message. The action tag “match” includes the control configuration of the smart TV 100B.

(109) Consider, Joy sends the reply message to Jessie. In response to receiving the reply message, the smartphone 200A detects the action tag “match” in the reply message as shown in notation (e) of the FIG. 11b. Further, the smartphone 200A determines the control configuration associated with the action tag. Further, the smartphone 200A determines that the smart TV 200B is unavailable to apply the control configuration. Further, the smartphone 200A determines that the smartphone 200A is relevant electronic device to play the video of cricket match. Further, the smartphone 200A converts the control configuration to the suitable control configuration to play the video of cricket match in the smartphone 200A. In response to receiving the user selection on the action tag from Jessie, the smartphone 200A applies the suitable control configuration to the smartphone 200A to play the video of cricket as shown in notation (f) of the FIG. 11b.

(110) The embodiments disclosed herein can be implemented using at least one software program running on at least one hardware device and performing network management functions to control the elements.

(111) The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the scope of the embodiments as described herein.

Claims

1. A method for predicting content based recommendations in an Internet of Things (IoT) environment, comprising: recognizing, by a first electronic device in a first IoT network, content displayed on a display of the first electronic device; displaying, on the display of the first electronic device, at least one recommendation corresponding to the recognized content based on a history of user activities performed by a user using at least one of the first electronic devices and other electronic devices in the first IoT network; detecting, by the first electronic device, a user input selecting the at least one recommendation; determining, by the first electronic device, control configuration information based on device properties of a second electronic device in a second IoT

network; and sending, by the first electronic device, a message including the selected at least one recommendation and the determined control configuration information to the second electronic device in the second IoT network, wherein the message includes the control configuration information for setting of the second electronic device in the second IoT network, wherein the control configuration information specifies an action to be performed at the second electronic device in response to the user input selecting the at least one recommendation, and wherein device properties of the second electronic device are different from device properties of the first electronic device.

2. The method as claimed in claim 1, further comprises: receiving, by a second electronic device in the second IoT network, the message comprising the content and the selected at least one recommendation from the first electronic device in the first IoT network; determining, by the second electronic device in the second IoT network, at least one second electronic device in the second IoT network, relevant to the selected at least one recommendation in the message; and automatically configuring, by the second electronic device in the second IoT network, at least one relevant second electronic device in the second IoT network based on the selected at least one recommendation.

3. The method as claimed in claim 1, wherein the at least one recommendation is transferrable to the other electronic devices in the first IoT network and at least one second electronic device in a second IoT network.

4. The method as claimed in claim 1, generating, by the first electronic device, the at least one recommendation corresponding to the content, comprises: extracting, by the first electronic device, actionable data items from the content displayed on the display; determining, by the first electronic device, at least one first electronic device in the first IoT network, relevant to the actionable data items; and generating, by the first electronic device, the at least one recommendation representing the content displayed on the display by fetching actions related to at least one relevant first electronic device in the first IoT network based on a timestamp of a user activity and at least one electronic device on which the user activity is performed.

5. The method as claimed in claim 4, wherein the actionable data items are displayed on the first electronic device in one of a different color, an icon and a tagged user interface, and wherein the at least one recommendation indicates an information of at least one user activity performed on the at least one of the first electronic devices in the first IoT network and a control configuration for at least one second electronic device in a second IoT network to perform the at least one user activity.

6. A system for predicting content based recommendations in an Internet of Things (IoT) environment, wherein the system comprising a first IoT network and a second IoT network in which: a first electronic device in the first IoT network is configured to: recognize content displayed on a display of the first electronic device, display at least one recommendation corresponding to the recognized content based on a history of user activities performed by a user using at least one of the first electronic device and other electronic devices in the first IoT network, on a display of the first electronic device, detect a user input for selecting the at least one recommendation, determine by the first electronic device, control configuration information based on device properties of a second electronic device in a second IoT network, and send a message including the selected at least one recommendation to a second IoT network, wherein the message includes control configuration information for setting of a second electronic device in the second IoT network, wherein the control configuration information specifies an action to be performed at the second electronic device in response to the user input selecting the at least one recommendation; and wherein device properties of the second electronic device are different from device properties of the first electronic device.

7. The system as claimed in claim 6, wherein the at least one recommendation is transferrable to the other first electronic devices and at least one second electronic device in the second IoT network.

8. The system as claimed in claim 6, generate the at least one recommendation corresponding to the content, comprises: extract actionable data items from the content displayed on the display; determine at least one other electronic device in the first IoT network which is relevant to the actionable data items; and generate the at least one recommendation representing the content displayed on the first electronic device by fetching actions related to at least one other electronic device in the first IoT network based on a timestamp of a user activity and at least one electronic device on which the user activity is performed.

9. The system as claimed in claim 8, wherein the actionable data items are displayed on the first electronic device in one of a different color, an icon and a tagged user interface, and wherein the at least one recommendation indicates an information of at least one user activity performed on the at least one other electronic device in the first IoT network and a control configuration for at least one second electronic device in the second IoT network to perform the at least one user activity.
